Assignment no 5:

Sharma Ji’s Canteen System

The Scenario

“Sharma Ji” runs a popular college canteen. During lunch hours, it becomes too busy to manage orders manually. He has hired you to build a Python program to manage billing and the kitchen workflow.

The Requirements

You must use **Data Structures (Lists, Tuples, Sets, Dictionaries)** and **Functions** to complete this assignment.

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1. Setup Data

Menu Dictionary

Create a dictionary named **menu** containing at least 5 food items and their prices.
Example:

- Samosa → 15
- Chai → 10
- Maggi → 30
- Sandwich → 25
- Coffee → 20

GST Tuple

Create a tuple named **gst** to store the tax rate.

- GST = 5% → `0.05`

- Remember: **Tuples are immutable (cannot be changed).**

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2. Function 1: Verify the Order → `verify_order()`

Input

A list of items ordered by a student.
Example:
`["Chai", "Burger", "Chai"]`

Logic

- "Burger" does not exist in the menu.
- You must remove items that are not present in the menu.

Output

Return a **clean list** that contains only valid menu items.

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3. Function 2: Kitchen Slip → `print_kitchen_slip()`

Input

The verified list of items.

Logic

The cook does not need duplicate entries.
For example: If Chai is ordered 10 times, the cook only needs to know that Chai must be prepared.

Task

- Convert the list into a **Set** to remove duplicates.
- Print the unique items.

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4. Function 3: Generate Bill → `generate_bill()`

Input

- Verified list of items
- Menu dictionary

Logic

1. Calculate the **total price** of all items.
2. Calculate **Tax** using the GST tuple value.
   `Tax = Total × GST`
3. Add tax to the total.

Output

Return the **final amount** the student has to pay.

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5. Special Task: Diwali Offer (Dictionary Comprehension)

Sharma Ji wants to give a festive discount.

Task

Use **Dictionary Comprehension** to create a new dictionary called **offer_menu**.

Rule

Reduce the price of **every item** in the original menu by **₹2**.

Final Execution (Main Section)

A student named **Rohit** places the following order:

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["Chai", "Samosa", "Chai", "Burger", "Maggi"]
```

Steps to Perform

1. Call **verify_order()** to clean the order.
2. Call **print_kitchen_slip()** to display items for the cook.
3. Use the **offer_menu** to calculate the final bill.
4. Call **generate_bill()** to print the final payable amount.